Unsupervised Spatiotemporal Segmentation & Flow Tracking of Fluid Dynamics

A 17-Step Mathematical Pipeline for 16-Bit High-Speed Imaging

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Primary Motivation & Core Objectives

Overarching Goal: Macro-scale fluid dynamics (e.g., shear force calculations) demand ultra-precise, sub-pixel edge tracking that standard static algorithms cannot provide.

The Challenges

1. Low-Contrast Artifacts & Segmentation Failure:
Low-contrast mist and specular reflections obscure true boundaries. Standard global binarization suffers from pixel-density bias, causing severe over-segmentation.

2. Fragmentation & Internal Voids:
Identifying the true jet structure is severely hindered by the presence of surrounding disconnected optical noise and empty internal structures (hollow voids).

Pipeline Objectives

1. Probabilistic Core Isolation:
Achieve unsupervised isolation using a frame-adaptive nested GMM to cleanly separate the dense fluid mass from the surrounding mist.

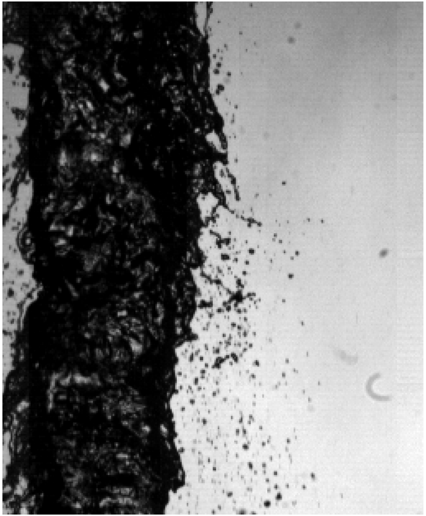
2. Graph-Based Shape Extraction:
Execute a graph-based search technique (Connected Component Labeling) to identify closely attached parts, bypass ambient noise, and reconstruct the true shape of the jet.

Step 1: Raw Frame Acquisition

- **Loads the raw 16-bit high-speed sensor data for the current temporal state.**
- *Formula:*

$$\boxed{I_t(i,j) \in [0, 65535] \quad \forall (i,j) \in \Omega}$$

Step 1: Raw Test Frame 150
(Before any processing)



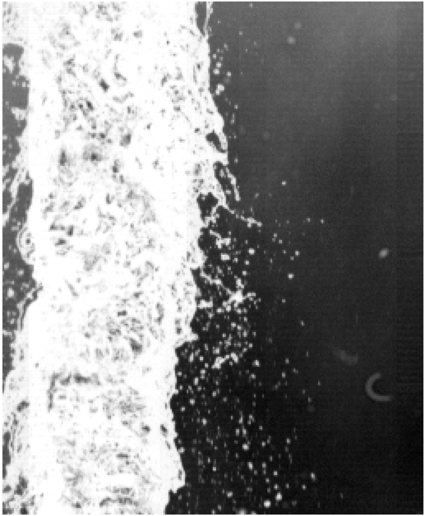
Step 2: Local Min-Max Inversion

- **Applies a strict 16-bit range inversion strictly relative to the localized physical peak of the current frame. This prevents global exposure drift and ensures dimensional stability across frames.**
- *Formula:*

$$\boxed{I^t_{max} = \max_{(i,j)\in\Omega} I_t(i,j)} \qquad \boxed{I^t_{min} = \min_{(i,j)\in\Omega} I_t(i,j)}$$

$$\boxed{I^t_{inv}(i,j) = \frac{I^t_{max} - I_t(i,j)}{I^t_{max} - I^t_{min} + 10^{-6}} \times 65535}$$

Step 2: Local Min-Max Inversion
((I_max - I_t) / (I_max - I_min)) * 65535



Step 3: Dynamic Nested GMM Processing

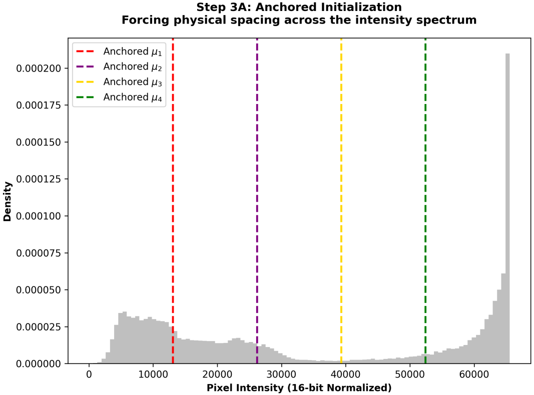
- **An unsupervised statistical breakdown of the frame's intensity spectrum.**

Visualization: https://codepen.io/Mradul-Namdeo/pen/bNgBWGZ

Step 3A: Initialization

- **Spreads four initial distributions evenly across the intensity range, forcing the algorithm to acknowledge all physical density states without bias**
- *Formula:*

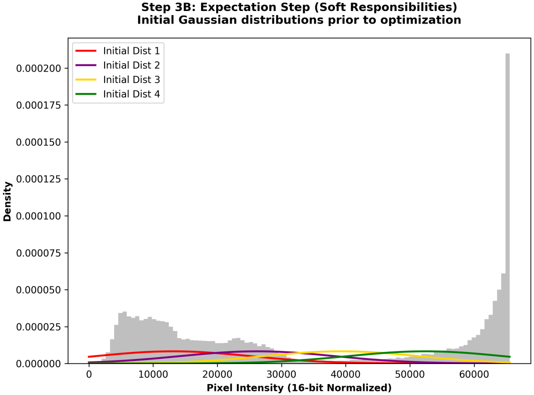
$$\boxed{\mu_{init}^{(k)} = I_{min}^t + \frac{k}{5}(I_{max}^t - I_{min}^t), \;\; k \in \{1..4\}}$$



Step 3B: Expectation (E-Step)

- **Calculates the membership probability (formally γ, or responsibility)—evaluating the exact mathematical odds that a specific pixel x_i belongs to density state k.**
- *Formula:*

$$\boxed{\gamma_{i,k} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{4} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}}$$

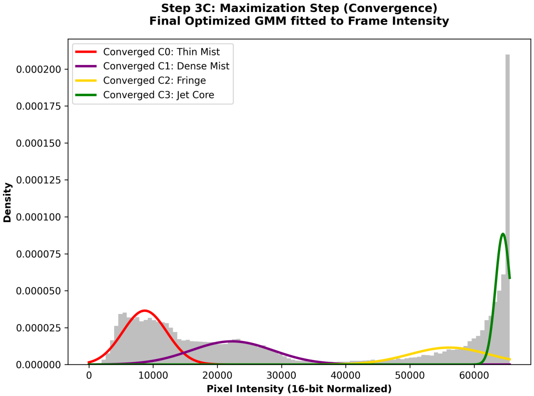


Step 3C: Maximization (M-Step)

- **Recomputes the expected center (μ) and adjusts the variance (σ^2) of each density state to perfectly fit the probabilistically assigned pixels. The algorithm sequentially iterates 3B and 3C until the model converges.**
- *Formula:* Let $\boxed{N_k = \sum_i \gamma_{i,k}}$

$$\boxed{\mu_k = \frac{1}{N_k} \sum_i \gamma_{i,k} x_i} \quad \text{(Center)}$$

$$\boxed{\sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{i,k} (x_i - \mu_k)^2} \quad \text{(Width)}$$



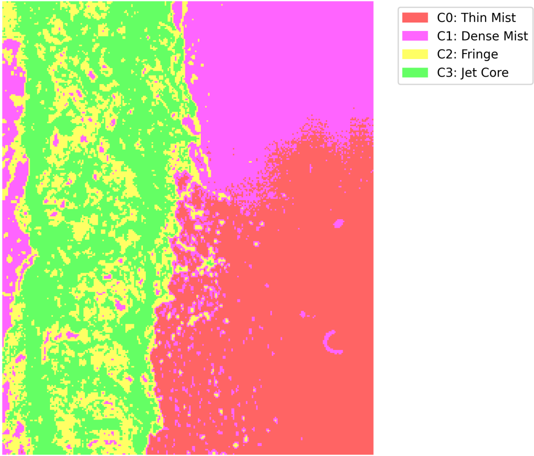
Step 3D: Cluster-2 (Fringe) Isolation

- **Sorts the converged distributions from darkest to brightest. Isolates the 3rd cluster (C_2), capturing the critical physical transition zone between mist and dense fluid.**
- *Formula:*

$$\boxed{\textit{Sorted} : \mu_1 < \mu_2 < \mu_3 < \mu_4}$$

$$\boxed{C_2 = \{x_i \mid \arg\max_k(\gamma_{i,k}) = 3\}}$$

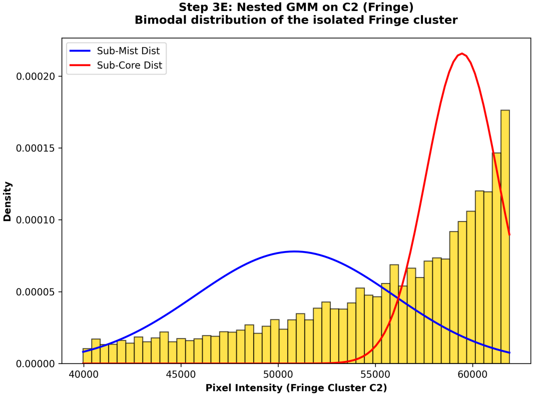
Step 3D: 2D GMM Hard Assignment Map (Mapped on Test Frame 150)



Step 3E: Hierarchical GMM

- **Applies a secondary, Hierarchical 2-component GMM exclusively onto the isolated C_2 cluster to split the transition zone into a lighter "sub-mist" and a denser "sub-core".**
- *Formula:*

$$P(x \mid C_2) = \sum_{m=1}^{2} \pi_{sub,m} \mathcal{N}(x \mid \mu_{sub,m}, \sigma^2_{sub,m})$$

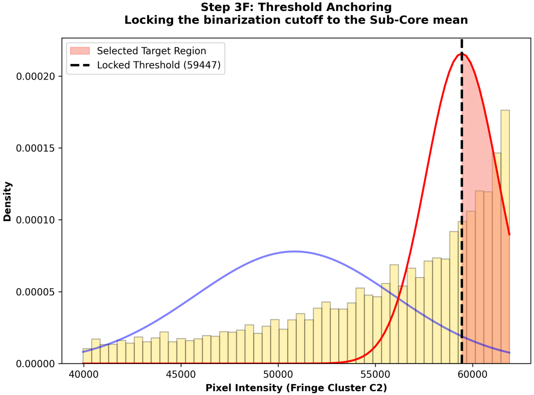


Step 3F: Threshold Criteria

- **Selects the upper sub-mean ($\mu_{sub,2}$), representing the denser sub-core, as the baseline criteria. A structural offset (δ_{shift}) is added to define the final rigid binarization threshold.**
- *Formula:*

$$Sorted\ Sub{-}Means : \boxed{\mu_{sub,1} < \mu_{sub,2}}$$

$$\boxed{\tau^{t}_{local} = \mu_{sub,2} + \delta_{shift}}$$



Step 4: Target Binarization

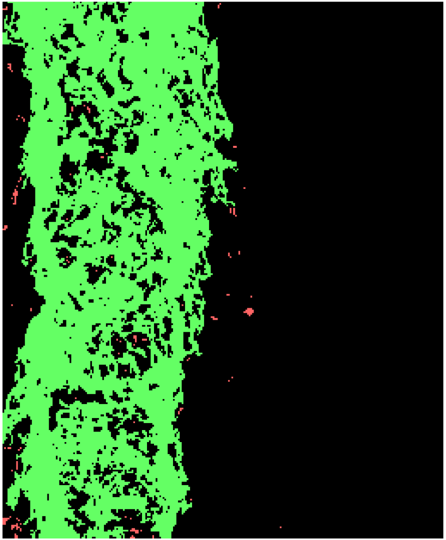
- **Evaluates every spatial pixel against the dynamically generated, highly precise Sub-Core cutoff.**
- *Formula:*

$$\boxed{B^t_{bin}(i,j) = \begin{cases} 255 & \text{if } I^t_{inv}(i,j) > \tau^t_{local} \\ 0 & \text{otherwise} \end{cases}}$$

Step 5A: Priority CCL & Noise Map

- **Extracts components using 8-connectivity.**

Step 5A: CCL Priority Isolation

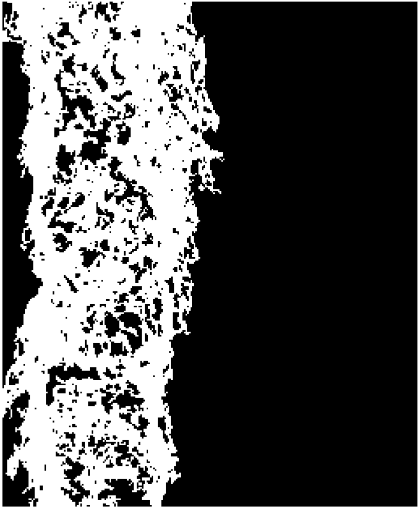


Step 5B: Priority CCL & Noise Map

- **Implements a Physical Constraint Prioritization:** The component set is searched iteratively to identify the largest component anchored at the source boundary ($Y = 0$).
- if no such component exists, the area-maximization prior is applied.
- Let $L = \{L_1, \ldots, L_n\}$ be components sorted by area $|L_k|$.
  Formula:

$$\boxed{B^t_{area} = \begin{cases} L_k & \text{where } k = \min\{i \mid Y_{top}(L_i) = 0\} \\ L_1 & \text{if } \forall i, Y_{top}(L_i) > 0 \end{cases}}$$

Step 5B: Max Area Isolated

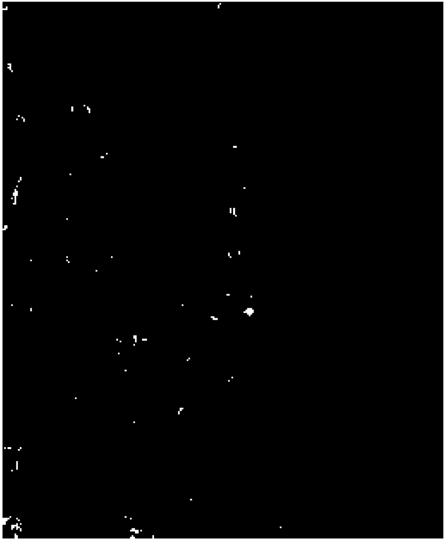


Step 5C: Priority CCL & Noise Map

- Following the identification of the dominant jet structure, B^t_{area}, the remaining components are treated as transient artifacts. We utilize the union operator, N^t_{map}, to consolidate these isolated pixel clusters into a collective **'Noise Map'**. This step is crucial, as it transforms fragmented, disconnected detections into a single map.
- *Formula:*

$$\boxed{N^t_{map} = \bigcup_{L_k \neq B^t_{area}} L_k]}$$

Step 5C: Noise Map



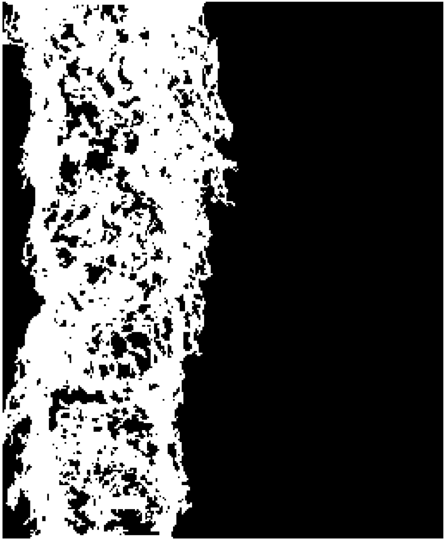
Step 6: Row-Specific Capping

- Extracts extreme X-coordinates exclusively at the absolute top and bottom Y-rows of the priority mass, creating a rigid mathematical cap strictly within the object's physical bounds to ensure perfect structural sealing.
- *Formula:*

$$\boxed{DrawLine : (X_{min}^{top}, Y_{top}) \leftrightarrow (X_{max}^{top}, Y_{top})}$$

$$\boxed{DrawLine : (X_{min}^{bot}, Y_{bot}) \leftrightarrow (X_{max}^{bot}, Y_{bot})}$$

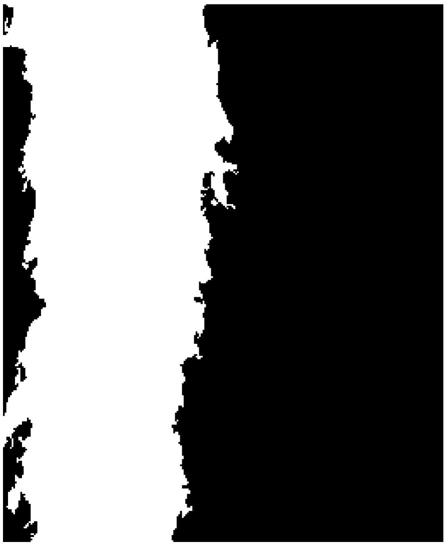
Step 6: Horizontal Capping



Step 7: Preliminary Solid Mask

- Resolves internal segmentation voids (e.g., central specular reflections and localized low-intensity gradients). Invokes the topological flood-fill operation (Jordan Curve Theorem) to isolate the closed contour of the capped mask, defining the solid fluid core as the union of the boundary and its bounded topological interior. Hence, converting the incomplete jet structure into a completely solid binary mask.
- Let int(X) denote the bounded topological interior of a closed contour X.

  Formula: $\boxed{M^t_{fill} = B^t_{cap} \cup \mathrm{int}(B^t_{cap})}$



Step 8: Post Masking CCL

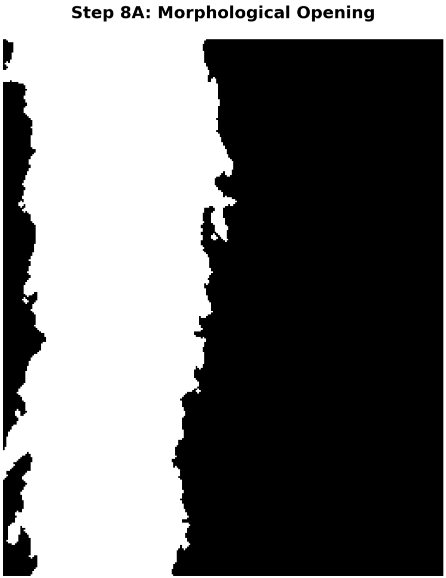
- Severs weak filament connections and re-evaluates the topology to securely isolate the dense geometric core.

Step 8A: Morphological Opening

- Applies a morphological opening operation using an elliptical structuring element (3 $\times$ 3) to detach transient artifacts and sever thin fluid filaments.
- Let $S_{3\times3}$ be an elliptical structuring element.

  Formula:

$$\boxed{M^t_{open} = M^t_{fill} \circ S_{3\times3}}$$



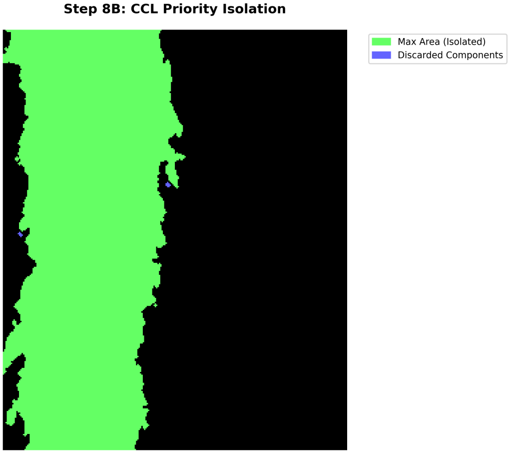
Step 8B: Topological Core Isolation

- Re-applies 8-way CCL to partition the opened mask into disconnected clusters (V), sorted by descending area to definitively extract the geometric fluid core and discard remaining noise.
- *Formula:*

$$\boxed{V = \mathrm{CCL}_8(M^t_{open}) = \{V_1, V_2, \dots, V_m\}}$$

Sorted Condition: $\boxed{|V_1| \geq |V_2| \geq \cdots \geq |V_m|}$

$$\boxed{B^t_{core} = \begin{cases} V_k & \text{where } k = \min\{i \mid Y_{top}(V_i) = 0\} \\ V_1 & \text{if } \forall i, Y_{top}(V_i) > 0 \end{cases}}$$

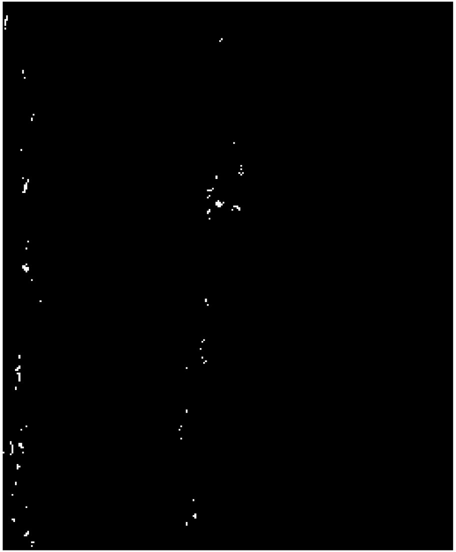


Step 9: Fragment Detachment

- Computes the absolute spatial deviation between the bulk filled mask and the isolated solid core to designate independent, detached mass candidates.
- *Formula:*

$$\boxed{C^t_{cands} = |M^t_{fill} - B^t_{core}|}$$

Step 9: Fragment Detachment



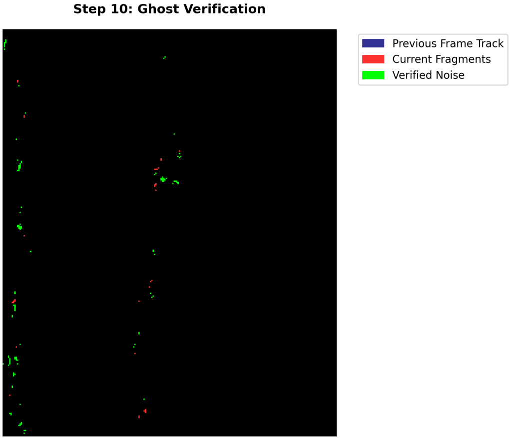
Step 10: Ghost Verification

- Interrogates detached fragments against historical boundaries. If a fragment heavily intersects the previous frame's core, or crosses the historical search track, it is verified as artifact noise rather than valid fluid mass.
- *Formula:* For each connected fragment $\boxed{f \in C^t_{cands}}$:

$$\boxed{C_1 : \frac{|f \cap B^{t-1}_{core}|}{|f|} < \theta_{overlap}} \quad \text{(where } \theta = 0.1\text{)}$$

$$\boxed{C_2 : |f \cap P^{t-1}| > 0}$$

$$\boxed{B^t_{frag} = \bigcup \{f \in C^t_{cands} \mid C_1 \lor C_2\}}$$

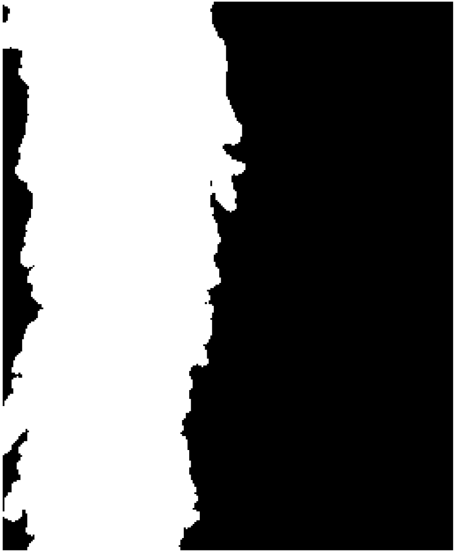


Step 11: Final Verified Mask

- Executes a Boolean NOT algebra to eliminate verified ghost fragments from the bulk mask, followed by a light morphological dilation to restore high-frequency physical perimeter data lost during processing.
- *Formula:*

$$\boxed{M^t_{final} = \left(M^t_{fill} \wedge \neg B^t_{frag}\right) \oplus S_{5\times5}}$$

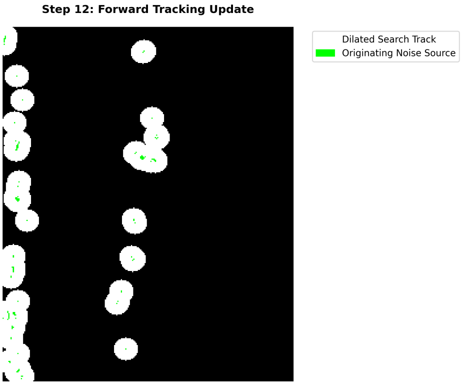
Step 11: The Final Mask



Step 12: Forward Tracking Update

- Heavily dilates the currently verified artifacts with a large elliptical kernel to generate a probability search track, transmitting spatial memory to the subsequent temporal frame.
- *Formula:*

$$\boxed{P^t = B^t_{frag} \oplus S_{21\times 21}}$$



Step 13: Topological Boundary Extraction

- Traces the continuous outer contour of the verified mask. Identifies the absolute right-most anchor points at the top and bottom bounds, splits the contour path in two, and retains exclusively the right-most topological edge for accurate curvature analytics.

Step 13: Topological Boundary Extraction

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- *Formula: Extract continuous contour path C. Then Split C at A_{top}, A_{bot} yielding paths P_1, P_2.*

$$\boxed{A_{top} = \arg\max_{x}\{x \mid (x, Y_{top}) \in C\}}$$

$$\boxed{A_{bot} = \arg\max_{x}\{x \mid (x, Y_{bot}) \in C\}}$$

$$\boxed{E^t_{right} = \begin{cases} P_1 & \text{if } \min(X \in P_1) > \min(X \in P_2) \\ P_2 & \text{otherwise} \end{cases}}$$

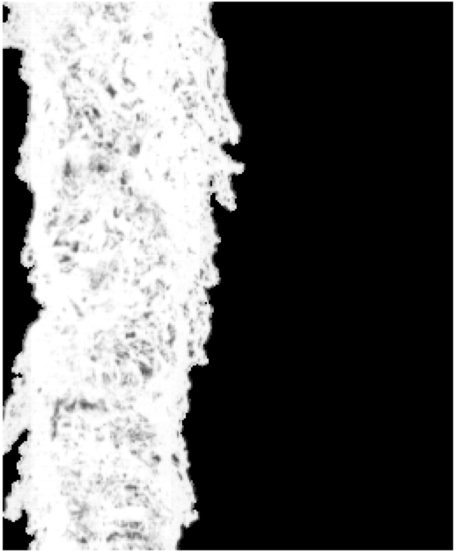
Step 14: Grayscale Target Extraction

- Maps the finalized mask directly onto the locally inverted original data. This completely nullifies the background while preserving the exact 16-bit physical intensities of the internal fluid dynamics.
- *Formula:*

$$\boxed{I^t_{target}(i,j) = \begin{cases} I^t_{inv}(i,j) & \text{if } M^t_{final}(i,j) > 0 \\ 0 & \text{otherwise} \end{cases}}$$

$$\boxed{\forall (i,j) \in \Omega}$$

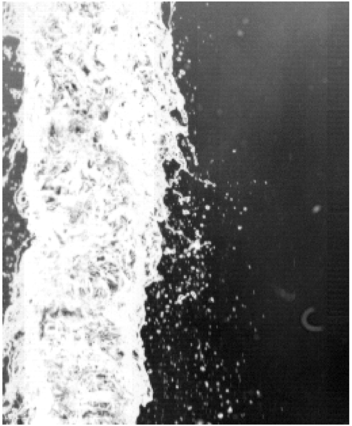
Step 14: Grayscale Target Extraction



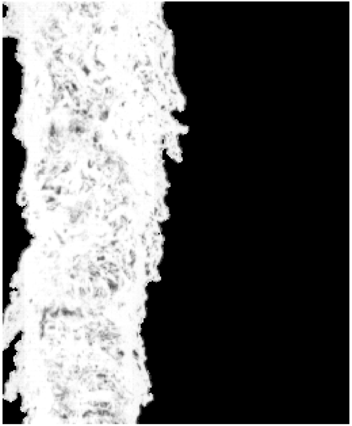
Final Result & Validation

- Comparison between Normalized raw image and Predicted jet of *"10kV ROI2 FV 75kfps 1p7slpm R1 C001H001S0001"*.

Step 2: Local Min-Max Inversion

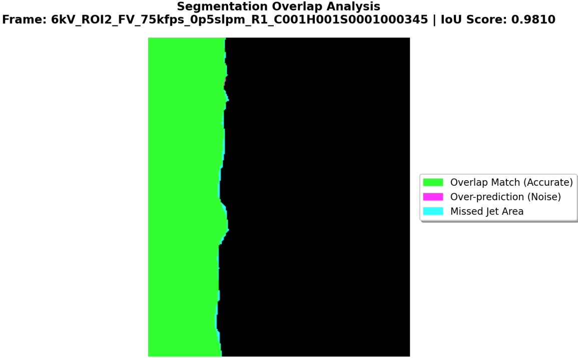


Step 14: Final Grayscale Target Extracted



Final Result & Validation

- Comparison between Manually annotated frame and Predicted frame by calculating the **Intersection over Union (IoU)**.
- IoU score for 6kV ROI2 FV 75kfps 0p5slpm R1 C001H001S0001000345 is 98%.



Conclusion

Summary of Achievements:

- **Algorithmic Robustness**: Successfully bypassed pixel-density bias by implementing localized, physical-spectrum anchored GMMs.
- **Spatiotemporal Tracking**: Engineered a dynamic pipeline capable of cross-referencing and eliminating persistent structural noise across high-speed video sequences.
- **Topological Precision**: Extracted mathematically pure boundary coordinates, isolating the absolute rightmost fluid interface while remaining immune to internal flow concavities.
- **Deep Learning Readiness**: Produced rigorously background-subtracted, locally normalized 16-bit inversions, establishing a pristine and standardized data pipeline for future deep learning feature extraction.

Research Papers

- A new image thresholding method based on Gaussian mixture model (ResearchGate).
- Spatially-Aware Gaussian Mixture Models for Structured Segmentation in Multispectral Satellite Imagery (IEEE).
- Sequential operations in digital picture processing (Rosenfeld Pfaltz) (ACM).
- Continuous crack detection using the combination of dynamic mode decomposition and connected component-based filtering method (ResearchGate).
- Efficient Spatio-temporal Segmentation for Extracting Moving Objects in Video Sequences (IEEE).
- A VOP Generation Tool: Automatic Segmentation of Moving Objects in Image Sequences Based on Spatio-Temporal Information (IEEE).
- Evaporation of a freely floating droplet in an airstream: effects of temperature, humidity and shape oscillations (JFM) (arXiv 2026).

Thank you!

Questions?